

CST8912 – Cloud Solution Architecture

Graded Lab Activity #10

Evaluate the security and data privacy implications of various cloud solutions.

Explore different encryption tools and technologies to protect the privacy and integrity of client's data

Describe authentication and identity management mechanisms to protect user's information

Identify best practices and industry standards to adopt a secure cloud to protect client information

Evaluate the levels of security for each cloud service model (IaaS, PaaS, SaaS)

Underline cloud security challenges and steps to maximize reliability and resiliency of application services

Introduction:

Azure Logic Apps is the PaaS (Platform as a Service) offering from Microsoft Azure. Logic Apps helps us to define workflows and build powerful solutions with the help of connectors, triggers, and actions.

Azure Logic Apps is a cloud platform where you can create and run automated workflows with little to no code. By using the visual designer and selecting from prebuilt operations, you can quickly build a workflow that integrates and manages your apps, data, services, and systems.

Few examples where logic apps can be used:

Schedule and send email notifications using Office 365 when a specific event happens, for example, a new file is uploaded.

Route and process customer orders across on-premises systems and cloud services.

Move uploaded files from an SFTP or FTP server to Azure Storage.

Monitor tweets, analyze the sentiment, and create alerts or tasks for items that need review.

Purpose of this lab:

Task 1: how you can alert users based on the data present/updated in your SQL Database.

Task 2: how to alert users when file gets uploaded/deleted in blob storage

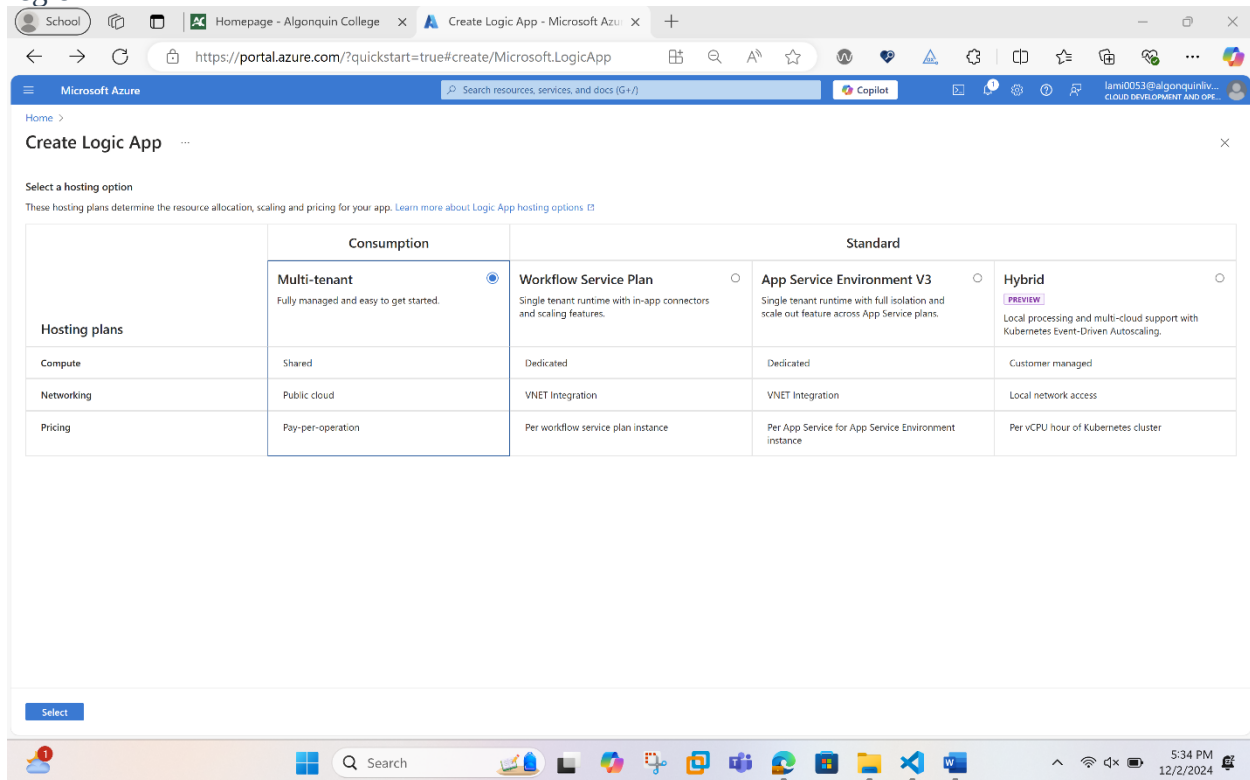
Task3 : Monitoring workflows in azure monitor

Task 4: Clean resources and record all the steps with screenshots in the lab report.

Problem statement: I have a storage account in Azure . In the container we are storing various data files . The files are stored in tree hierarchy of folders (Parent-> year -> month -> day).Each day new files get uploaded to the specific day folder . If the file for that specific day is not uploaded I would like to drop email notification

Task 1:

Create logic app (choose consumption based plan) and sql database instance in Canada central region



Home > Create Logic App ...

Select a hosting option

These hosting plans determine the resource allocation, scaling and pricing for your app. [Learn more about Logic App hosting options](#)

	Consumption		Standard	
Hosting plans	Multi-tenant ☒ Fully managed and easy to get started.	Workflow Service Plan ☐ Single tenant runtime with in-app connectors and scaling features.	App Service Environment V3 ☐ Single tenant runtime with full isolation and scale out feature across App Service plans.	Hybrid ☐ PREVIEW Local processing and multi-cloud support with Kubernetes Event-Driven Autoscaling.
Compute	Shared	Dedicated	Dedicated	Customer managed
Networking	Public cloud	VNET Integration	VNET Integration	Local network access
Pricing	Pay-per-operation	Per workflow service plan instance	Per App Service for App Service Environment instance	Per vCPU hour of Kubernetes cluster

Select

School

Homepage - Algonquin College

Create Logic App (Multi-tenant)

https://portal.azure.com/?quickstart=true#view/WebsitesExtension/WorkflowApp...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

lami0053@algonquiniv...
CLOUD DEVELOPMENT AND OPS...

Home > Create Logic App >

Create Logic App (Multi-tenant)

Basics Tags Review + create

Create a logic app, which lets you group workflows as a logical unit for easier management, deployment and sharing of resources. Workflows let you connect your business-critical apps and services with Azure Logic Apps, automating your workflows without writing a single line of code.

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance Details

Logic App name *

Region *

Enable log analytics * ☐ Yes ☒ No

There are no log analytics workspace resources in the selected subscription. In order to enable log analytics, either create a new log analytics workspace resource or switch to a subscription which already has one.

Looking for the classic consumption create experience? [Click here](#)

[Review + create](#) [< Previous](#) [Next: Tags >](#)

School

Homepage - Algonquin College

Create SQL Database - Microsoft

https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

lami0053@algonquiniv...
CLOUD DEVELOPMENT AND OPS...

Home > SQL databases >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Want to try Azure SQL Database for free? Create a free serverless database with the first 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of the subscription. [Learn more](#)

[Apply offer \(Preview\)](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server *

[Review + create](#) [Next: Networking >](#)

The screenshot displays the Microsoft Entra ID console interface during the 'Create SQL Database Server' wizard. The wizard is currently at the 'Authentication' step. The 'Server name' is set to 'lab10server1', and the 'Location' is '(Canada) Canada Central'. Under 'Authentication method', the option 'Use both SQL and Microsoft Entra authentication' is selected. In the 'Set Microsoft Entra admin' section, 'Not Selected' is indicated, with a 'Set admin' link. The 'Server admin login' field contains 'Enter server admin login', the 'Password' is masked with asterisks, and the 'Confirm password' field is empty. On the right, the 'Selected item' pane shows 'Keshav Lamichhane' with the email 'kiam0053@algonquiniv...'. The bottom of the screen shows the Windows taskbar with the Start menu, search bar, and various application icons.

The screenshot shows the 'Create SQL Database Server' wizard in the Microsoft Azure portal. The 'Authentication' step is selected, showing options for Microsoft Entra ID authentication. The 'Server admin login' is set to 'Not Selected'. The 'Server admin login' field is empty. The 'Password' field is masked with dots. The 'Confirm password' field is empty. The 'OK' button is visible at the bottom left.

School

Homepage - Algonquin College

Create SQL Database Server - Mic

[←](#)
[↻](#)
<https://portal.azure.com/?quickstart=true#view/SqlAzureExtension/SharedCreate...>

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lami0053@algonquinliv...
CLOUD DEVELOPMENT AND OPE...

[Home](#) > [SQL databases](#) > [Create SQL Database](#) >

Create SQL Database Server

Microsoft

Server name *

lab10server1

.database.windows.net

Location *

(Canada) Canada Central

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) , or select both SQL and Microsoft Entra authentication.

Authentication method

☐ Use Microsoft Entra-only authentication
 ☒ Use both SQL and Microsoft Entra authentication
 ☐ Use SQL authentication

Set Microsoft Entra admin

lami0053@algonquinlive.com

Admin Object/App ID: 8663620a-ee5f-453e-b790-4a6420bf9354

[Set admin](#)

Server admin login *

adminserver

Password *

Confirm password *

OK

School

Homepage - Algonquin College

Create SQL Database - Microsoft

[←](#)
[↻](#)
<https://portal.azure.com/?quickstart=true#create/Microsoft.SQLDatabase>

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lami0053@algonquinliv...
CLOUD DEVELOPMENT AND OPE...

[Home](#) > [SQL databases](#) >

Create SQL Database

Microsoft

[Basics](#)
[Networking](#)
[Security](#)
[Additional settings](#)
[Tags](#)
[Review + create](#)

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'lab10server1' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

☐ No access
 ☒ Public endpoint
 ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Connection policy

Configure how clients communicate with your SQL database server. [Learn more](#)

Connection policy

☒ Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure
 ☐ Proxy - All connections are proxied via the Azure SQL Database gateways
 ☐ Redirect - Clients establish connections directly to the node hosting the

Review + create

< Previous

Next : Security >

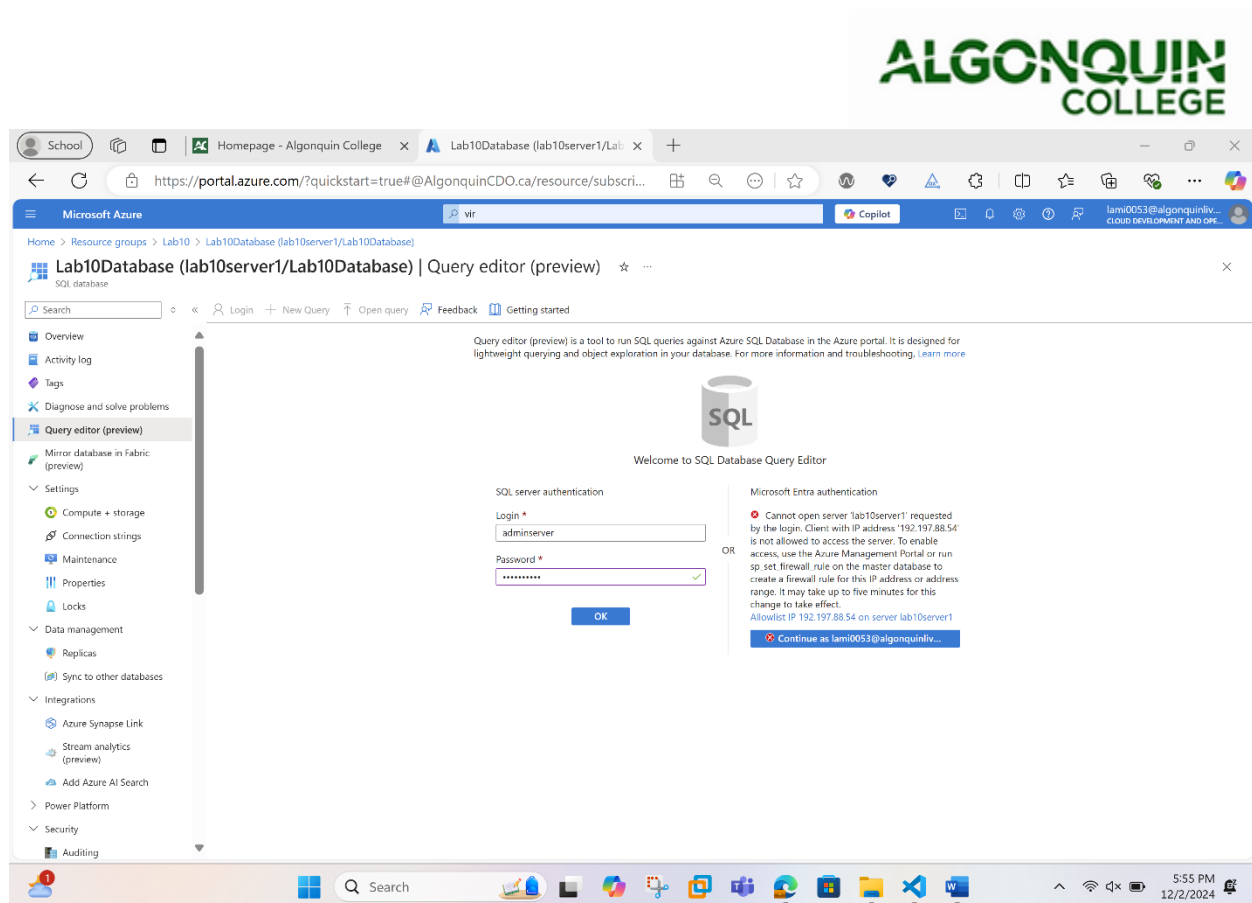
SQL

Cost summary

General Purpose (GP_S_Gen5_1)	
Cost per GB (in USD)	0.13
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	5.26 USD
COMPUTE COST / VCORE SECOND *	0.000174 USD

NOTES

* Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)



Create alerts table in sql database using query editor

```
create table Alerts
(
  ID int NOT NULL PRIMARY KEY IDENTITY (1,1),
  ToAddress varchar(50) NOT NULL,
  MailSubject varchar(50) NOT NULL,
  MailBody varchar(50) NOT NULL,
  EmailSent bit NOT NULL
)
GO
```

School | Homepage - Algonquin College | Lab10Database (lab10server1/Lab10Database) | SQL Table Creation

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/subscri...

Microsoft Azure | vir | Copilot | lami0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Resource groups > Lab10 > Lab10Database (lab10server1/Lab10Database)

Lab10Database (lab10server1/Lab10Database) | Query editor (preview)

SQL database

Search | Login | New Query | Open query | Feedback | Getting started

Lab10Database (adminserver)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

- Tables
- Views
- Stored Procedures

Query 1

Run | Cancel query | Save query | Export data as | Show only Editor | Open Copilot

```

1 CREATE TABLE Alerts
2 (
3     ID INT NOT NULL PRIMARY KEY IDENTITY(1,1),
4     ToAddress VARCHAR(50) NOT NULL,
5     MailSubject VARCHAR(50) NOT NULL,
6     MailBody VARCHAR(50) NOT NULL,
7     EmailSent BIT NOT NULL
8 )
9 GO
10

```

Results | Messages

Search to filter items...

Ready

School | Homepage - Algonquin College | Lab10Database (lab10server1/Lab10Database) | SQL Table Creation

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/subscri...

Microsoft Azure | vir | Copilot | lami0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Resource groups > Lab10 > Lab10Database (lab10server1/Lab10Database)

Lab10Database (lab10server1/Lab10Database) | Query editor (preview)

SQL database

Search | Login | New Query | Open query | Feedback | Getting started

Lab10Database (adminserver)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

- Tables
- Views
- Stored Procedures

Query 1

Run | Cancel query | Save query | Export data as | Show only Editor | Open Copilot

```

1 CREATE TABLE Alerts
2 (
3     ID INT NOT NULL PRIMARY KEY IDENTITY(1,1),
4     ToAddress VARCHAR(50) NOT NULL,
5     MailSubject VARCHAR(50) NOT NULL,
6     MailBody VARCHAR(50) NOT NULL,
7     EmailSent BIT NOT NULL
8 )
9 GO
10

```

Results | Messages

Query succeeded: Affected rows: 0

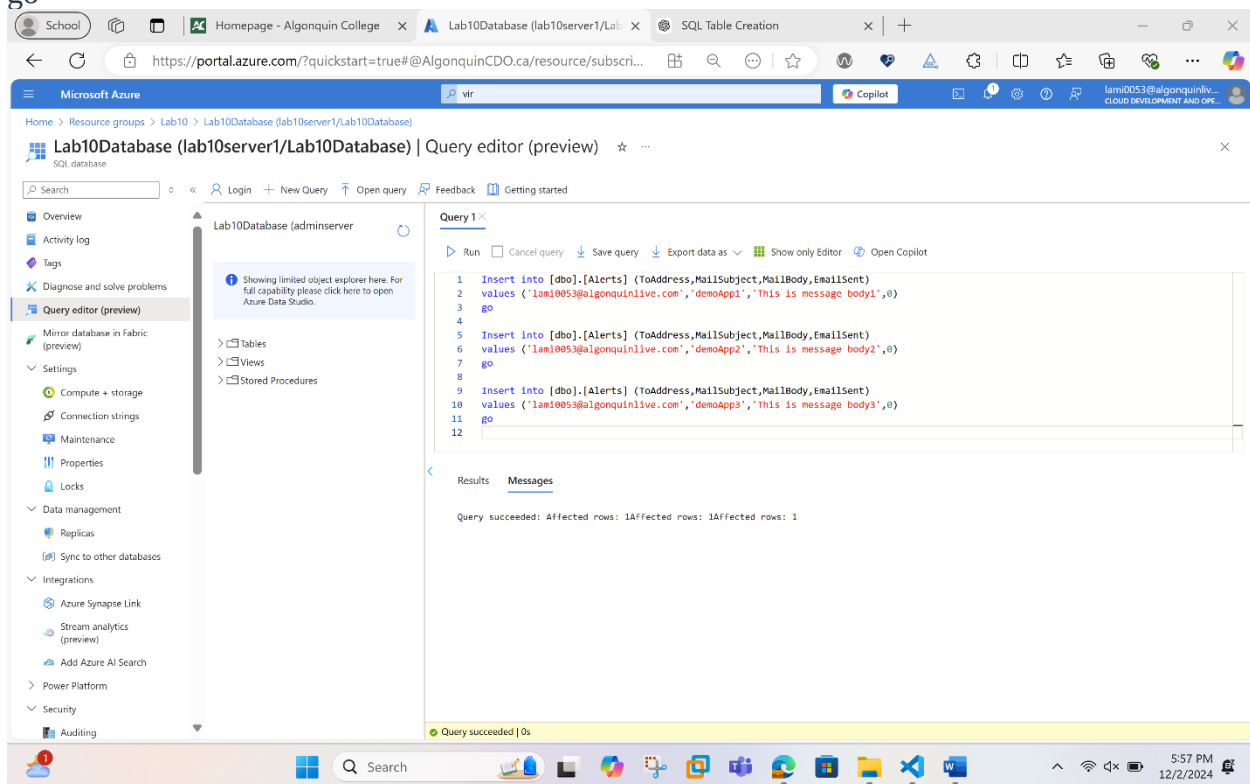
Query succeeded | 0s

Insert records into the Alerts table using the query

```
Insert into [dbo].[Alerts] (ToAddress,MailSubject,MailBody,EmailSent)
values ('youremail','demoApp1','This is message body1',0)
go
```

```
Insert into [dbo].[Alerts] (ToAddress,MailSubject,MailBody,EmailSent)
values ('youremail','demoApp2','This is message body2',0)
go
```

```
Insert into [dbo].[Alerts] (ToAddress,MailSubject,MailBody,EmailSent)
values ('youremail','demoApp3','This is message body3',0)
go
```



Select rows from db.Alerts table to verify the records inserted in the table.



School

Homepage - Algonquin College

Lab10Database (lab10server1/Lab10Database)

SQL Table Creation

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/subscri...

Microsoft Azure

vir

Copilot

lami0053@algonquinlive...
CLOUD DEVELOPMENT AND OPE...

Home > Resource groups > Lab10 > Lab10Database (lab10server1/Lab10Database)

Lab10Database (lab10server1/Lab10Database) | Query editor (preview)

SQL database

Search

« Login + New Query Open query Feedback Getting started

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Settings

Compute + storage

Connection strings

Maintenance

Properties

Locks

Data management

Replicas

Sync to other databases

Integrations

Azure Synapse Link

Stream analytics (preview)

Add Azure AI Search

Power Platform

Security

Auditing

Lab10Database (adminserver)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

dbo.Alerts

Views

Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor Open Copilot

1 select * from [dbo].[Alerts]

Results Messages

Search to filter items...

ID	ToAddress	MailSubject	MailBody	EmailSent
1	lami0053@algonquinlive.com	demoApp1	This is message body1	False
2	lami0053@algonquinlive.com	demoApp2	This is message body2	False
3	lami0053@algonquinlive.com	demoApp3	This is message body3	False

Query succeeded | 0s

1

Search



5:58 PM
12/2/2024

Go to logic app created in lab

Use recurrence trigger and define values for interval (3) and frequency (minute)

School | Homepage - Algonquin College | Logicapp - Microsoft Azure

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/subscri...

Microsoft Azure | Search | Add a trigger

Home > Resource groups > Lab10 > Logicapp

Logicapp | Logic app designer

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Development Tools | Logic app designer | Logic app code view | Run history | Versions | API connections | Quick start guides | Settings | Workflow settings | Authorization | Access keys | Identity | Properties | Locks | Monitoring | Alerts | Metrics | Diagnostic settings

Add a trigger

recurrence

Runtime: Select a runtime | Action type: Triggers

Group by Connector

Loading more results...

Schedule (In-app) | Recurrence (In-app) | Amazon SQS | CRM Bot | Bttn ONE | lbttnWebhook

Search resources, services, and docs (G+/I)

Logic apps | Add | Manage view

Logicapp | Logic app designer

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Development Tools | Logic app designer | Logic app code view | Run history | Versions | API connections | Quick start guides | Settings | Workflow settings | Authorization | Access keys | Identity | Properties | Locks | Monitoring | Alerts | Metrics | Diagnostic settings

Recurrence

Parameters | Settings | Code view | About

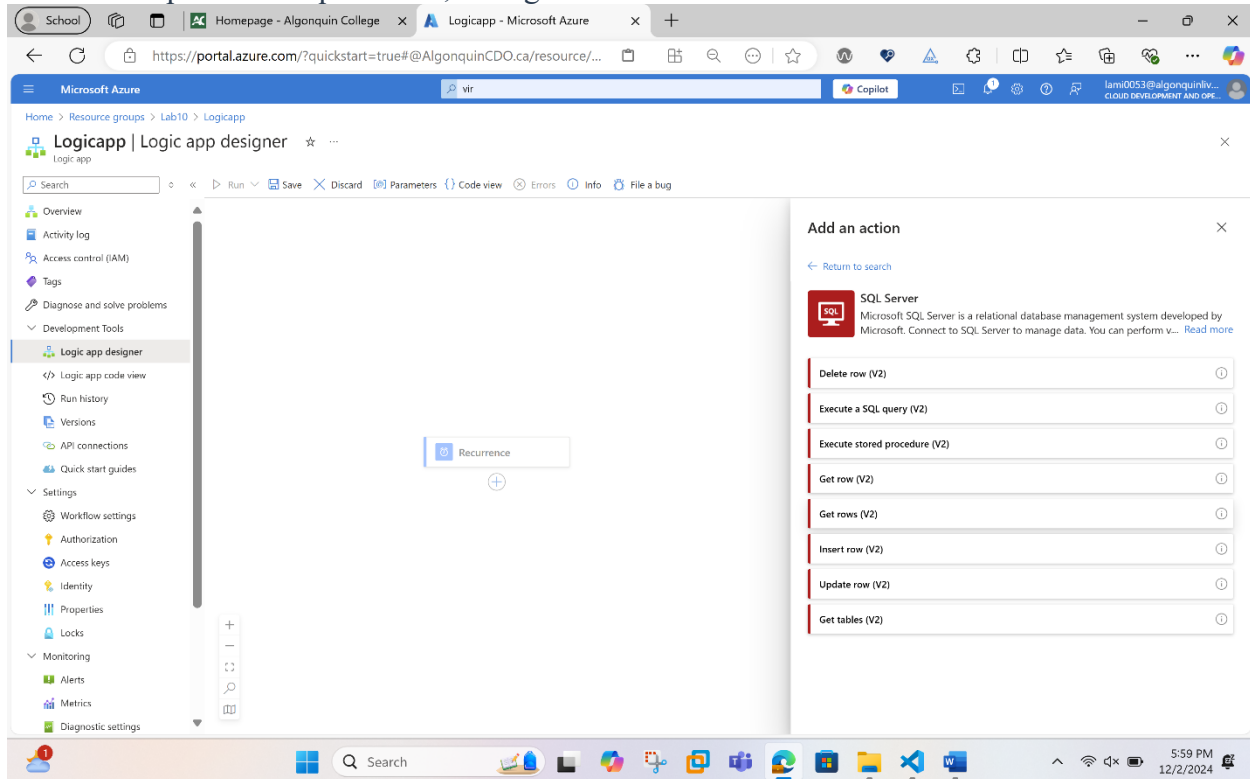
Recurrence *

Interval: 3 | Frequency: Minute

Time Zone: Select timezone

Start Time: Example: 2017-03-24T15:00:00Z

Add new step named “sql server”, use “get rows” as action.



Enter the credentials (In background, connectors are getting created). Enter your server name (FQDN), database name, username and password.

Microsoft Azure | Logic app designer

Home > Resource groups > Lab10 > Logicapp

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Development Tools
Logic app designer
Logic app code view
Run history
Versions
API connections
Quick start guides
Settings
Workflow settings
Authorization
Access keys
Identity
Properties
Locks
Monitoring
Alerts
Metrics
Diagnostic settings

Recurrence

Get rows (V2)

Create connection

Get rows (V2)

Create a new connection

Connection Name *
Authentication Type *
SQL Server Name *
SQL Database Name *
Username *
Password *
Gateway

Subscription
Gateway

Create new

Microsoft Azure | lab10server1 | Networking

Home > lab10server1

Search

Overview
Activity log
Access control (IAM)
Tags
Quick start
Diagnose and solve problems
Settings
Microsoft Entra ID
SQL databases
SQL elastic pools
Properties
Locks
Data management
Security
Networking
Microsoft Defender for Cloud
Transparent data encryption
Identity
Auditing
Intelligent performance
Automatic tuning
Recommendations

Public network access

Public Endpoints allow access to this resource through the internet using a public IP address. An application or resource that is granted access with the following network rules still requires proper authorization to access this resource. [Learn more](#)

Public network access

☐ Disable
☒ Selected networks

Connections from the IP addresses configured in the Firewall rules section below will have access to this database. By default, no public IP addresses are allowed. [Learn more](#)

Virtual networks

Allow virtual networks to connect to your resource using service endpoints. [Learn more](#)

+ Add a virtual network rule

Rule	Virtual network	Subnet	Address range	Endpoint status	Resource group	Subscription	State
No virtual network rules are currently configured.							

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (192.197.88.54) + Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address
Access	13.71.175.160	13.71.175.160
query-editor-9c9f32	192.197.88.54	192.197.88.54

Exceptions

Save Discard

ALGONQUIN
COLLEGE

School | Homepage - Algonquin College | Logicapp - Microsoft Azure | lab10server1 - Microsoft Azure | +

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/...

Microsoft Azure | Search | Run | Save | Discard | Parameters | Code view | Errors | Info | File a bug

Logic app designer

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Development Tools | Logic app designer | Logic app code view | Run history | Versions | API connections | Quick start guides | Settings | Workflow settings | Authorization | Access keys | Identity | Properties | Locks | Monitoring | Alerts | Metrics | Diagnostic settings

Recurrence

Get rows (V2)

Parameters | Settings | Code view | Testing | About

Server Name *
Use connection settings (lab10server1.database.windows.net)

Database Name *
Use connection settings (Lab10Database)

Table Name *
Alerts

Advanced parameters
Showing 0 of 6 | Show all | Clear all

Connected to Sqlconnection. | Change connection

6:05 PM 12/2/2024

Add a new step 'For-Each' in the Logic App.
Select 'Value' Parameter from Dynamic Content

The screenshot displays the Microsoft Azure Logic App Designer interface. The workflow is titled 'Logic app' and is located in the 'Lab10' resource group. The workflow consists of the following steps:

- Recurrence** (Trigger)
- Get rows (V2)** (Action)
- For each** (Loop)

The 'For each' loop is currently selected, and its 'Parameters' tab is active on the right-hand pane. The 'Parameters' tab shows a selection of 'value' from the previous step's output. The left-hand navigation pane shows various settings and monitoring options. The top of the browser window shows the URL 'https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/...' and the user 'lam0053@algonquiniv...'. The bottom of the screen shows the Windows taskbar with the time '6:06 PM 12/2/2024'.

Now add a 'Send mail' Action.
Enter the details from Dynamic content (refer to values from columns defined in the alerts table)

School | Homepage - Algonquin College | Logicapp - Microsoft Azure | lab10server1 - Microsoft Azure

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/...

Microsoft Azure | Logicapp | Logic app designer

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Development Tools | Logic app designer | Logic app code view | Run history | Versions | API connections | Quick start guides | Settings | Workflow settings | Authorization | Access keys | Identity | Properties | Locks | Monitoring | Alerts | Metrics | Diagnostic settings

Recurrence

Get rows (V2)

For each

Send an email (V2)

Create connection

Send an email (V2)

Create a new connection

Sign in to create a connection to Office 365 Outlook.

Sign in

School | Homepage - Algonquin College | Logicapp - Microsoft Azure | lab10server1 - Microsoft Azure

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/...

Microsoft Azure | Logicapp | Logic app designer

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Development Tools | Logic app designer | Logic app code view | Run history | Versions | API connections | Quick start guides | Settings | Workflow settings | Authorization | Access keys | Identity | Properties | Locks | Monitoring | Alerts | Metrics | Diagnostic settings

Recurrence

Get rows (V2)

For each

Send an email (V2)

Send an email (V2)

Parameters

To *

ToAddress x

Subject *

MailSubject x

Body *

Normal Arial 15px B I U A

MailBody x

Advanced parameters

Showing 1 of 7 Show all Clear all

Importance

Normal

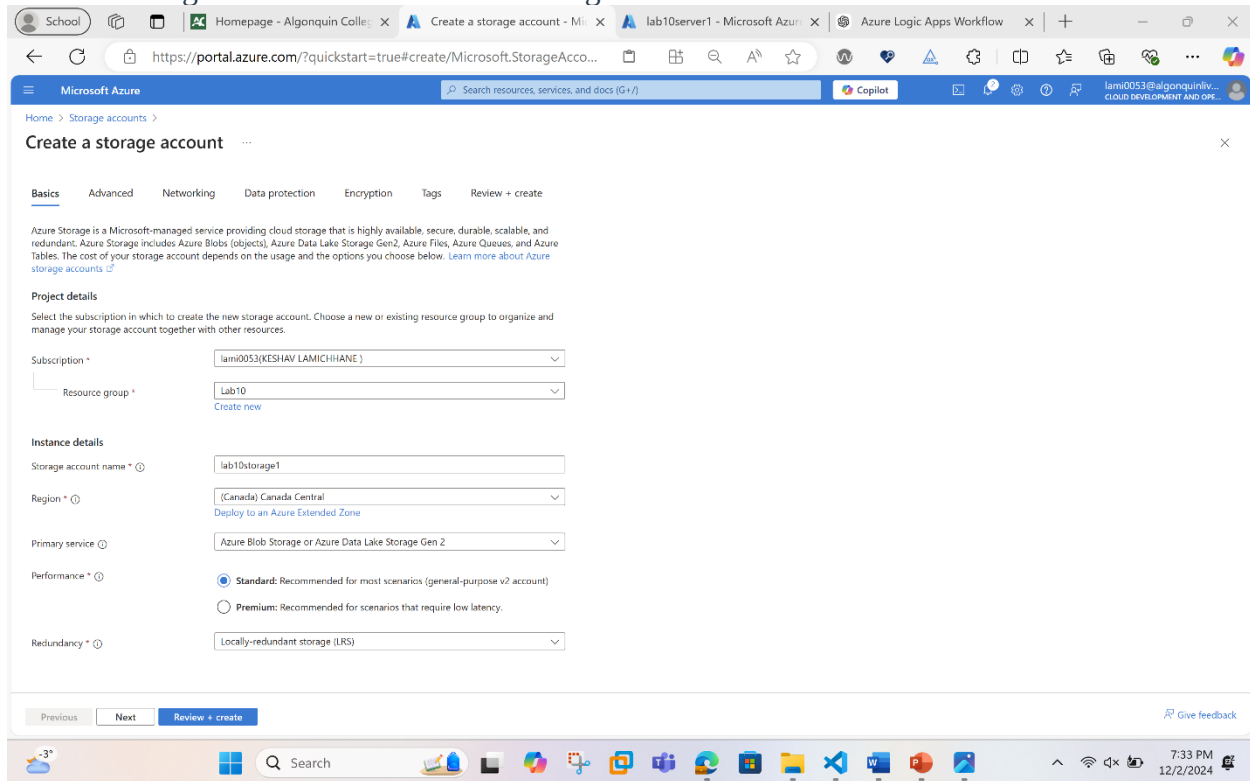
Connected to lami0053@algonquinlive.com. Change connection

Save the Logic App.
Wait for sometime and you will receive an email.

The screenshot displays the Microsoft Outlook web interface in a browser window. The browser's address bar shows the URL <https://outlook.office.com/mail/>. The Outlook interface includes a left sidebar with navigation links for Favorites (Inbox, Sent Items, Drafts) and Folders (Inbox, Drafts, Sent Items, Deleted Items, Junk Email, Archive, Notes, Conversation History, Search Folders). The main content area is divided into two panes: a list of emails and a detailed view of the selected email. The email list shows several messages from Keshav Lamichhane, each with a subject line (demoApp1, demoApp2, demoApp3) and a body preview (This is message body1, This is message body2, This is message body3). The selected email, 'demoApp3', is shown in the right pane, displaying the sender's name, email address, and the body text. The interface also includes a top navigation bar with 'Home', 'View', and 'Help' tabs, and a bottom taskbar with various application icons.

Task 2: Design a logic to trigger an email notification in your outlook when the file to a specific folder does not gets uploaded by specific time.

Create a storage account in Canada central region



The screenshot shows the Microsoft Azure portal interface for creating a new storage account. The browser tabs include 'School', 'Homepage - Algonquin College', 'Create a storage account - Microsoft Azure', 'lab10server1 - Microsoft Azure', and 'Azure Logic Apps Workflow'. The URL is [https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount...](https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount). The user is logged in as 'lami0053@algonquiniv...'. The 'Create a storage account' wizard is displayed with the 'Basics' tab selected. The 'Project details' section shows the subscription 'lami0053(KESHAV LAMICHIANE)' and resource group 'Lab10'. The 'Instance details' section shows the storage account name 'lab10storage1', region '(Canada) Canada Central', and primary service 'Azure Blob Storage or Azure Data Lake Storage Gen 2'. The 'Performance' section has 'Standard' selected, and the 'Redundancy' section has 'Locally-redundant storage (LRS)' selected. The bottom navigation bar shows 'Previous', 'Next', and 'Review + create' buttons. The system tray at the bottom shows the date and time as 7:33 PM on 12/2/2024.

Create a sample container and within that container create folder in format of “yyyy-mm-dd”

The screenshot shows the Microsoft Azure portal interface. The main pane displays the 'container1' storage container with an 'Overview' tab selected. The 'Upload blob' dialog is open on the right, showing a file named 'High level diagram.png' selected for upload. The dialog includes options for 'Block blob' type, '4 MiB' block size, and 'Hot (Inferred)' access tier. The 'Upload to folder' field is set to '2024/12/02'. The 'Encryption scope' is set to 'Use existing default container scope'.

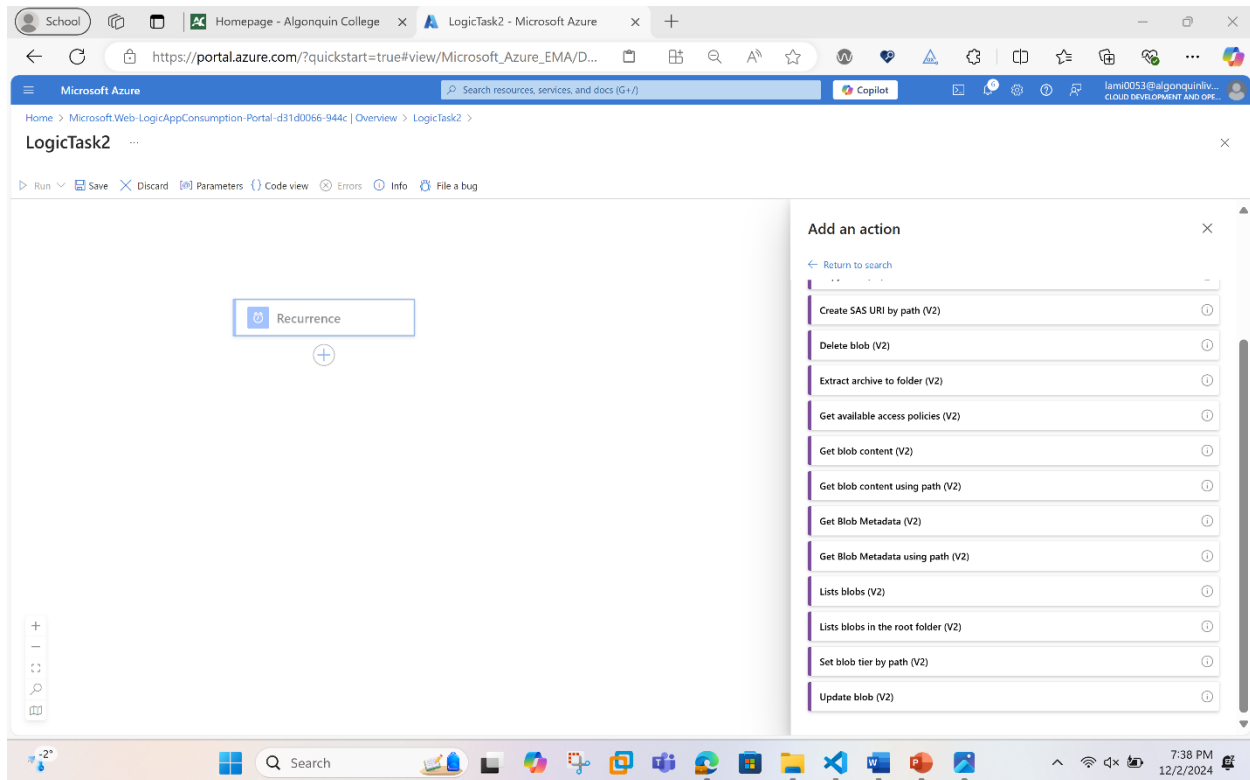
Create a trigger to schedule this logic everyday at 6pm

The screenshot shows the Microsoft Azure Logic App Designer interface. The main pane displays a logic app named 'LogicTask2' with a 'Recurrence' trigger. The 'Recurrence' trigger is configured with the following settings:

- Interval:** 1
- Frequency:** Day
- Time Zone:** (UTC -05:00) Eastern Time (US & Canada)
- Start Time:** 16:00:00
- At These Hours:** Example: 0, 10
- At These Minutes:** Enter the valid minute values (from 0 to 59) separated by comma, e.g., 15,30

The 'Preview' section shows 'Runs every day'.

Use list blob to check every file in folder to check the file in the folder, you can use the expression like `concat('/',utcNow('yyyy/MM/dd'))`, if the file in the path does not exists, list blob will fail.



School | Homepage - Algonquin College | LogicTask2 - Microsoft Azure | lab10storage1 - Microsoft Azure

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_EMA/D...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Microsoft.Web-LogicAppConsumption-Portal-d31d0066-944c | Overview > LogicTask2 >

LogicTask2

Run | Save | Discard | Parameters | Code view | Errors | Info | File a bug

```

graph TD
    Recurrence[Recurrence] --> Lists[Lists blobs (V2)]
  
```

Create connection

Lists blobs (V2)

Create a new connection

Connection Name *

Authentication Type *

Azure Storage Account Name Or Blob Endpoint *

Azure Storage Account Access Key

Create new

School | Homepage - Algonquin College | LogicTask2 - Microsoft Azure | lab10storage1 - Microsoft Azure

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_EMA/D...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Microsoft.Web-LogicAppConsumption-Portal-d31d0066-944c | Overview > LogicTask2 >

LogicTask2

Run | Save | Discard | Parameters | Code view | Errors | Info | File a bug

```

graph TD
    Recurrence[Recurrence] --> Lists[Lists]
  
```

Lists

Parameters | Settings | Code view | Testing | About

Storage Account Name Or Blob Endpoint *

Folder *

Advanced parameters
Showing 2 of 2

Paging Marker
A marker that identifies the portion of the list to be returned with the list opera...

Flat Listing

Connected to connection2 [Change connection](#)

School | Homepage - Algonquin College | LogicTask2 - Microsoft Azure | lab10storage1 - Microsoft Azure | (8) WhatsApp

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_EMA/D...

Microsoft Azure | Search resources, services, and docs (G+V) | Copilot | lam0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Microsoft.Web-LogicAppConsumption-Portal-d31d0066-944c | Overview > LogicTask2 >

LogicTask2

Run | Save | Discard | Parameters | Code view | Errors | Info | File a bug

The workflow diagram shows a Recurrence trigger connected to a Lists action, which is then connected to a Condition action. The Condition action is currently being edited, showing a search for 'Function' and a list of string functions including concat, substring, slice, replace, and guid.

Condition Expression *

Provide the values to compare and select the operator to use.

Choose a value | is equal to | Choose a value

+ New item

Function | Dynamic content

Search

String functions

- concat(text_1, text_2, ...)
- Concatenates any number of strings together
- substring(text, startIndex, length?)
- Returns a subset of characters from a string.
- slice(text, startIndex, endIndex?)
- Returns a section of a string defined by the start index and the en...
- replace(text, oldText, newText)
- Replaces a string with a given string
- guid()
- Generates a globally unique string (GUID)

Add

School | Homepage - Algonquin College | LogicTask2 - Microsoft Azure | lab10storage1 - Microsoft Azure | (8) WhatsApp

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_EMA/D...

Microsoft Azure | Search resources, services, and docs (G+V) | Copilot | lam0053@algonquiniv... CLOUD DEVELOPMENT AND OPE...

Home > Microsoft.Web-LogicAppConsumption-Portal-d31d0066-944c | Overview > LogicTask2 >

LogicTask2

Run | Save | Discard | Parameters | Code view | Errors | Info | File a bug

The workflow diagram shows a Recurrence trigger connected to a Lists action, which is then connected to a Condition action. The Condition action is now fully configured with the expression 'empty(body('Lists')['value'])' and is set to 'is equal to' 'true'.

Condition Expression *

Provide the values to compare and select the operator to use.

AND

empty(body('Lists')['value']) | is equal to | true

+ New item

Microsoft Azure

LogicTask2

Run Save Discard Parameters Code view Errors Info File a bug

Send an email (V2)

Parameters Settings Code view Testing About

To *

lami0053@algonquinlive.com

Subject *

NO File Found

Body *

Please Upload File

Advanced parameters

Showing 1 of 7 Show all Clear all

Importance

Normal

Connected to lami0053@algonquinlive.com. Change connection

Microsoft Azure

LogicTask2

Run Save Discard Parameters Code view Errors Info File a bug

Send an email (V2) 1

Parameters Settings Code view Testing About

To *

lami0053@algonquinlive.com

Subject *

Success

Body *

You have a file in a folder.

Advanced parameters

Showing 1 of 7 Show all Clear all

Importance

Normal

Connected to lami0053@algonquinlive.com. Change connection

Send an email to your email address in outlook if list blob fails.

The screenshot shows the Microsoft Outlook web interface. The left sidebar displays the 'Favorites' and 'Folders' sections. The main pane shows a list of emails from 'Keshav Lamichhane' with subject lines like 'Success', 'demoApp3', 'demoApp2', 'demoApp1', and 'demoApp3'. The selected email is from 'Keshav Lamichhane' with the subject 'Success' and the body text 'You have a file in a folder.' The right pane shows the details of the selected email, including the sender's name, email address, and the body text. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 7:46 PM on 12/2/2024.

Task 3: Monitor workflows in azure logic apps.

The screenshot shows the Microsoft Azure portal interface. The left sidebar displays the 'Logic app' section with options like 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Diagnose and solve problems', 'Development Tools', 'Settings', and 'Monitoring'. The main pane shows the 'Run history' for a Logic app named 'LogicTask2'. The 'Run history' table displays the following data:

Status	Start time	Identifier	Duration	Static Results
Succeeded	12/2/2024, 7:46 PM	08584684201076875991705293973CU13	738 Milliseconds	

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 7:46 PM on 12/2/2024.

The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar shows the user is logged in as 'lami0053@algonquiniv...' with the role 'CLOUD DEVELOPMENT AND OPE...'. The main content area is titled 'LogicTask2 | Run history' and shows a table of run history with columns for 'Start time' and 'Duration'. A single run is listed with a start time of '12/2/2024, 7:46 PM' and a duration of '738 Milliseconds'. To the right, the 'LogicTask2' workflow diagram is shown, featuring a 'Recurrence' trigger, a 'Lists' action, a 'Condition' action, and two 'Send an email (V2)' actions branching from the condition.

LogicTask2 Run history

Start time	Duration
12/2/2024, 7:46 PM	738 Milliseconds

LogicTask2 Workflow Diagram

```

graph TD
    Recurrence[Recurrence] --> Lists[Lists]
    Lists --> Condition[Condition]
    Condition -- True --> SendEmailV2[Send an email V2]
    Condition -- False --> SendEmailV2_1[Send an email V2 1]
  
```

Task4: Clean all the resources created during this lab and record all the steps with screenshots in the lab report.

School | Homepage - Algonquin | Delete a resource group | lab10storage1 - Micro | (8) WhatsApp | Mail - Keshav Lamich |

https://portal.azure.com/?quickstart=true#@AlgonquinCDO.ca/resource/...

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Resource groups >

Resource groups

Cloud Development and Operations

+ Create Manage view

Filter for any field...

Name ↑

Lab10

Lab10

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Locks

Cost Management

Cost analysis

Cost alerts (preview)

Budgets

Advisor recommendations

Monitoring

Insights (preview)

Alerts

Metrics

Lab10

Subscription (copy) : lam0053/KESHAV LAMICHHANE

Subscription ID : b296b604-9c48-4e98-bb66-56c661c39a1d

Tags (edit) : Add tags

Resources

Recommendations (1)

Filter for any field... Type equals all Location equals all

Showing 1 to 8 of 8 records. Show hidden types

Name ↑

- azureblob
- Lab10Database (lab10server1/Lab10Database)
- lab10server1
- lab10storage1
- Logicapp
- LogicTask2
- office365
- sql

< Previous Page 1 of 1 Next >

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

Lab10

Dependent resources to be deleted (9)

All dependent resources, including hidden types, are shown

Name	Resource type
azureblob	API Connection
Lab10Database (lab10server1/Lab10Database)	SQL database
lab10server1	SQL server
lab10storage1	Storage account
Logicapp	Logic app
LogicTask2	Logic app
master (lab10server1/master)	SQL database
office365	API Connection
sql	API Connection

Enter resource group name to confirm deletion *

Lab10

Delete Cancel

School | Homepage - Algonquin | Home - Microsoft Azure | lab10storage1 - Micro | (8) WhatsApp | Mail - Keshav Lamich |

https://portal.azure.com/?quickstart=true#home

Microsoft Azure | Search resources, services, and docs (G+/I)

Home > Azure services >

Azure services

Create a resource

Resource groups

Logic apps

Storage accounts

SQL databases

Deploy a custom...

Virtual machines

Cost analysis

Service Bus

More services

Resources

Recent Favorite

Name	Type	Last Viewed
LogicTask2	Logic app	2 minutes ago
lab10storage1	Storage account	25 minutes ago
Logicapp	Logic app	2 hours ago
Lab10Database (lab10server1/Lab10Database)	SQL database	2 hours ago
lab10server1	SQL server	2 hours ago
lam0053/KESHAV LAMICHHANE	Subscription	2 weeks ago

See all

Navigate

Subscriptions

Resource groups

All resources

Dashboard

Tools

Notifications

More events in the activity log → Dismiss all

- Deleted resource group Lab10

Deleted resource group Lab10

a few seconds ago
- Successfully checked the trigger

Successfully checked the trigger of logic app 'LogicTask2' for new data

13 minutes ago
- Successfully saved workflow

Successfully saved workflow 'LogicTask2'

13 minutes ago
- Deployment succeeded

Deployment 'Microsoft.Web-LogicAppConsumption Portal-d3140066-944c' to resource group 'lab10' was successful.

Go to resource Go to resource group

23 minutes ago
- Successfully uploaded blob(s)

Successfully uploaded 1 blob(s).

24 minutes ago
- Successfully created storage container

Successfully created storage container 'container1'.

25 minutes ago
- Deployment succeeded

Deployment 'lab10storage1_1733186004465' to resource group 'Lab10' was successful.

Go to resource Pin to dashboard

